```
% ========================================================================
% Flat-Type Supercapacitor PDE Model with 2D Voltage & Energy Maps
% ========================================================================
clear; clc; close all;

%% 1. PARAMETERS
h_cap = 0.1;       % Height of cell [m]
l_a = 0.02;        % Anode thickness [m]
l_e = 0.02;        % Electrolyte thickness [m]
l_c = 0.02;        % Cathode thickness [m]

sigma_s = 1e-3;    % Electrode conductivity [S/m]
sigma_l = 1e-1;    % Electrolyte conductivity [S/m]

eps_s = 0.75;      % Electrode porosity
eps_l = 0.6;       % Electrolyte porosity

C_dl = 0.13;       % Double-layer capacitance [F/m^2]
A_v = 1.2e6;       % Specific surface area [1/m]

I_cell = 0.01;     % Applied cell current [A]

%% 2. CREATE PDE MODEL (Scalar PDE)
model = createpde(1);

%% 3. GEOMETRY (Flat Rectangular Layers, corrected)
% Rectangle format: [3;4;x1;x2;x3;x4;y1;y2;y3;y4]

% Anode
r1 = [3;4;0;l_a;l_a;0;0;0;h_cap;h_cap];

% Electrolyte
r2 = [3;4;l_a;l_a+l_e;l_a+l_e;l_a;0;0;h_cap;h_cap];

% Cathode
r3 = [3;4;l_a+l_e;l_a+l_e+l_c;l_a+l_e+l_c;l_a+l_e;0;0;h_cap;h_cap];

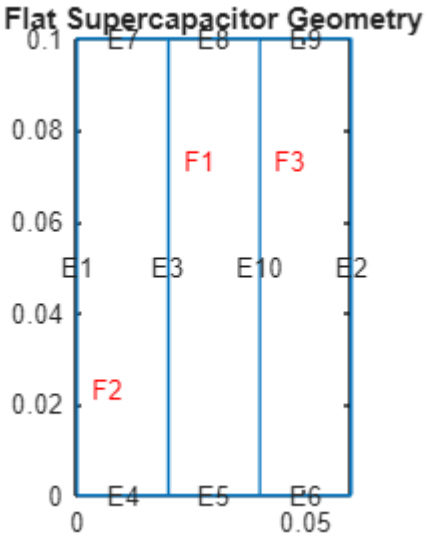
% Combine into geometry matrix
gd = [r1,r2,r3];

ns = char('Anode','Electrolyte','Cathode');
sf = 'Anode+Electrolyte+Cathode';

[g,~] = decsg(gd,sf,ns');
geometryFromEdges(model,g);

figure;
pdegplot(model,'EdgeLabels','on','FaceLabels','on');
axis equal;
```

```
title('Flat Supercapacitor Geometry');
```




```
%% 4. PDE COEFFICIENTS
% No axisymmetric scaling in flat capacitor
specifyCoefficients(model,'m',0,'d',@assignD,'c',@assignC,'a',0,'f',0);

%% 5. BOUNDARY CONDITIONS
% Cathode grounded (Dirichlet)
applyBoundaryCondition(model,'dirichlet','Edge',6,'u',0);

% Anode current input (Neumann)
applyBoundaryCondition(model,'neumann','Edge',1,'g',I_cell/h_cap); % Current per
unit height

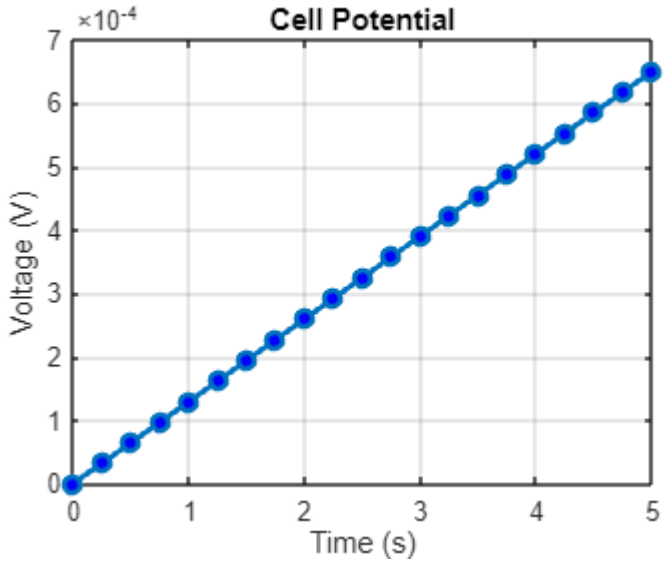
%% 6. MESH
generateMesh(model,'Hmax',h_cap/20);

%% 7. INITIAL CONDITIONS
setInitialConditions(model,1e-6);

%% 8. SOLVE TRANSIENT
tlist = 0:0.25:5;
results = solvepde(model,tlist);
u = results.NodalSolution;

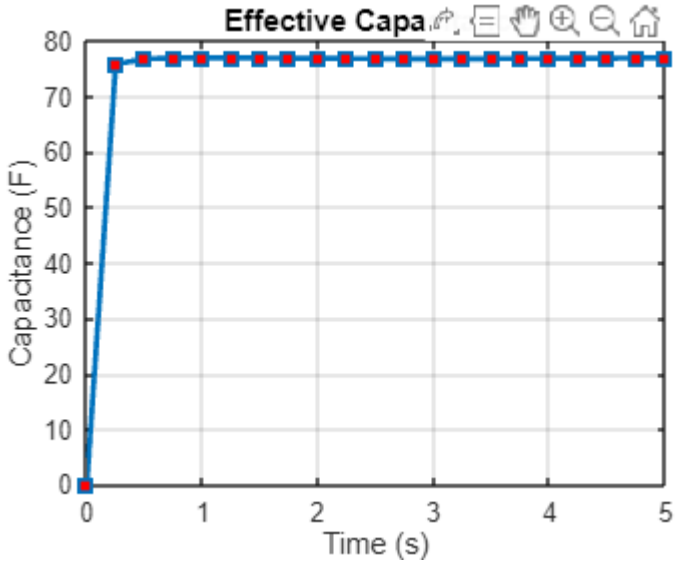
%% 9. POST-PROCESSING
% ---- Cell potential at anode (Face 1) ----
anode_nodes = findNodes(model.Mesh,'region','Face',1);
phi_anode = mean(u(anode_nodes,:),1);

figure;
plot(tlist,phi_anode,'-o','LineWidth',2,'MarkerFaceColor','b');
grid on; xlabel('Time (s)'); ylabel('Voltage (V)'); title('Cell Potential');
```

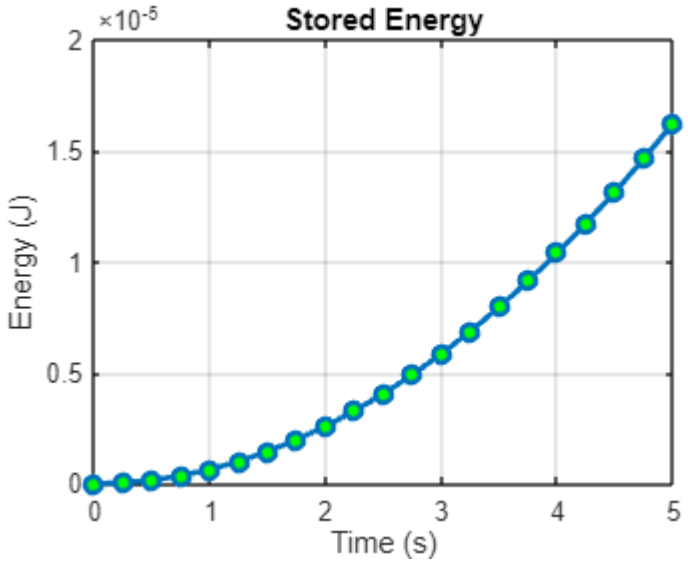



```
% ---- Effective capacitance ----
Q = cumtrapz(tlist,I_cell*ones(size(tlist)));
C_eff = Q ./ phi_anode;

figure;
plot(tlist,C_eff,'-s','LineWidth',2,'MarkerFaceColor','r');
grid on; xlabel('Time (s)'); ylabel('Capacitance (F)'); title('Effective
Capacitance');
```




```
% ---- Stored energy ----
E_stored = 0.5 * C_eff .* phi_anode.^2;
figure;
plot(tlist,E_stored,'-o','LineWidth',2,'MarkerFaceColor','g');
grid on; xlabel('Time (s)'); ylabel('Energy (J)'); title('Stored Energy');
```




```
fprintf('Final capacitance: %.4f F\n',C_eff(end));
```

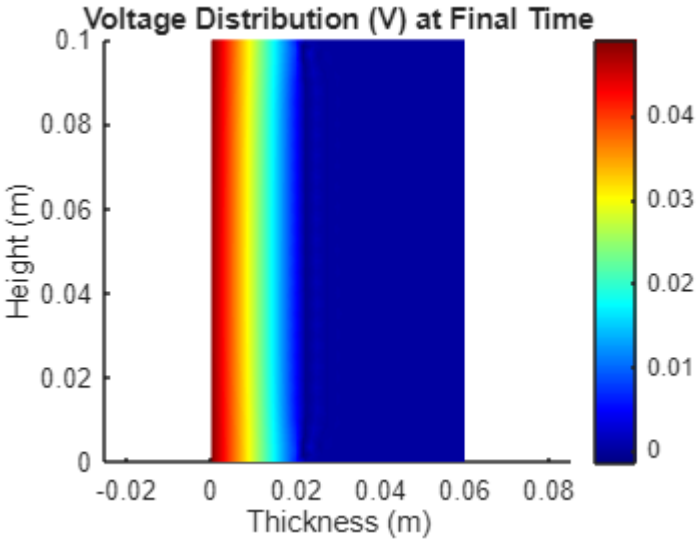
```
Final capacitance: 76.8991 F
```

```
fprintf('Final stored energy: %.4f J\n',E_stored(end));
```

```
Final stored energy: 0.0000 J
```

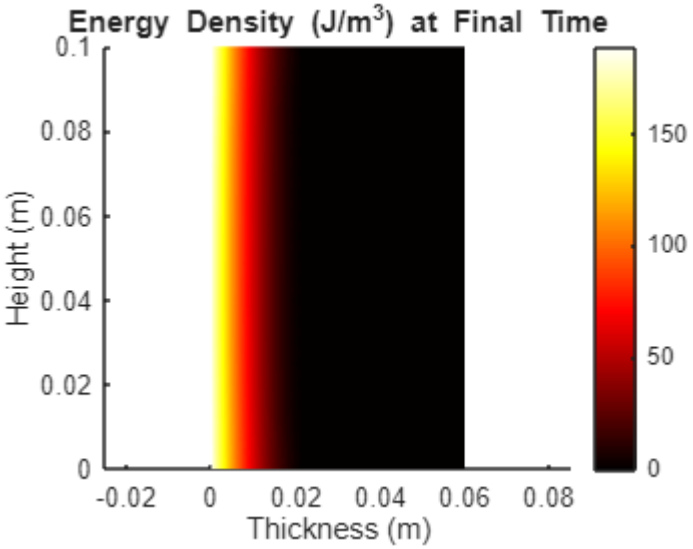
```
%% ---- 2D Voltage and Energy Density Maps (Final Time) ----
phi_final = u(:,end);

% Voltage distribution
figure;
pdeplot(model,'XYData',phi_final,'Mesh','off','ColorMap','jet');
axis equal; colorbar;
title('Voltage Distribution (V) at Final Time');
xlabel('Thickness (m)'); ylabel('Height (m)');
```




```
% Energy density distribution
C_dl_map = zeros(size(phi_final));
x_nodes = model.Mesh.Nodes(1,:);
C_dl_map(x_nodes <= l_a) = C_dl*A_v;                    % Anode
C_dl_map(x_nodes > l_a+l_e) = C_dl*A_v;                % Cathode
C_dl_map((x_nodes > l_a) & (x_nodes <= l_a+l_e)) = 1e-12; % Electrolyte

E_density = 0.5 * C_dl_map .* phi_final.^2;
figure;
pdeplot(model,'XYData',E_density,'Mesh','off','ColorMap','hot');
axis equal; colorbar;
title('Energy Density (J/m^3) at Final Time');
xlabel('Thickness (m)'); ylabel('Height (m)');
```



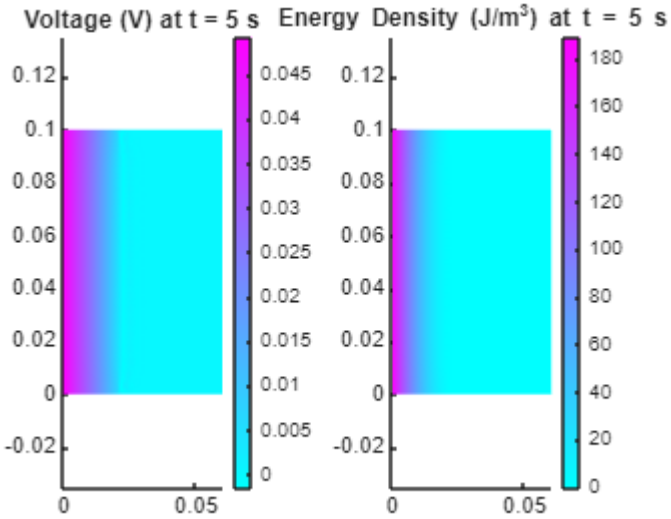

```
%% ---- Optional Animation Over Time ----
```

```
figure;
for k = 1:length(tlist)
    phi = u(:,k);
    E = 0.5*C_dl_map.*phi.^2;

    subplot(1,2,1);
    pdeplot(model,'XYData',phi,'Mesh','off'); axis equal; colorbar;
    title(['Voltage (V) at t = ', num2str(tlist(k)),' s']);

    subplot(1,2,2);
    pdeplot(model,'XYData',E,'Mesh','off'); axis equal; colorbar;
    title(['Energy Density (J/m^3) at t = ', num2str(tlist(k)),' s']);

    drawnow;
end
```




```
%% ====== Helper functions ======
function d = assignD(region,~)
    n = numel(region.x);
    d = zeros(1,n);

    isAnode = region.subdomain==1;
    isElectrolyte = region.subdomain==2;
    isCathode = region.subdomain==3;

    d(isAnode | isCathode) = 0.13*1.2e6;  % C_dl*A_v
    d(isElectrolyte) = 1e-12;             % small for solver
end

function c = assignC(region,~)
    n = numel(region.x);
    c = zeros(1,n);
```

```
    sigma_s = 1e-3; sigma_l = 1e-1;
    eps_s = 0.75; eps_l = 0.6;

    isAnode = region.subdomain==1;
    isElectrolyte = region.subdomain==2;
    isCathode = region.subdomain==3;

    c(isAnode | isCathode) = sigma_s * eps_s^1.5;
    c(isElectrolyte) = sigma_l * eps_l^1.5;
end
```